Quiz: Continuous Distributions (Advanced Level)

Time: 45 Minutes **Total Marks: 20**

Instructions: Attempt all questions. Show complete reasoning in proofs.

1

Let $X \sim U(a, b)$.

(a) If $P(X > 5) = \frac{1}{3}$ and $a = 2$, find b.

(b) A machine produces rods with lengths uniformly distributed between 10 cm and 20 cm. What is the probability that a randomly selected rod has length between 12 cm and 17 cm?

(c) Prove that the moment generating function (MGF) of X is:

$$M_X(t) = \frac{e^{tb} - e^{ta}}{t(b-a)}, \quad t \neq 0.$$

2

Let $X \sim \text{Exp}(\lambda)$.

(a) Given $P(X > 2) = 0.1353$, find λ.

(b) The lifetime of a device follows an exponential distribution with mean 5 hours. Find the probability that the device lasts more than 8 hours.

(c) Prove the memoryless property:

$$P(X > s + t \mid X > s) = P(X > t).$$

3

Let $X \sim \text{Gamma}(\alpha, \lambda)$.

(a) If $X_1 \sim \text{Exp}(\lambda)$ and $X_2 \sim \text{Exp}(\lambda)$ are independent, identify the distribution of $X_1 + X_2$.

(b) Suppose the waiting time for 3 events follows a Gamma distribution with $\alpha = 3$ and $\lambda = 2$. Find $E[X]$ and $\text{Var}(X)$.

(c) Prove that the MGF of Gamma distribution is:

$$M_X(t) = \left(\frac{\lambda}{\lambda - t}\right)^{\alpha}, \quad t < \lambda.$$

4

Let $X \sim \text{Beta}(\alpha, \beta)$.

(a) For $\alpha = \beta$, describe the symmetry of the distribution.

(b) A random variable $X \sim \text{Beta}(2, 3)$ models the proportion of defective items. Find $E[X]$.

(c) Prove that:

$$E[X] = \frac{\alpha}{\alpha + \beta}.$$

(Hint: Use Beta function properties.)

5

Let $X \sim N(\mu, \sigma^2)$.

(a) If $Z = \frac{X-\mu}{\sigma}$, show that $Z \sim N(0, 1)$.

(b) Heights of students are normally distributed with mean 170 cm and standard deviation 10 cm. Find the probability that a student's height is greater than 180 cm.

(c) Prove that the MGF of normal distribution is:

$$M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right).$$